

Aim

To write a Java program that demonstrates the ArrayList class methods add(), get(), set(), remove(), contains(), size(), isEmpty(), clear(), indexOf(), lastIndexOf() and sort().

Algorithm

1. Start.
2. Import the java.util package.
3. Create an ArrayList<String> object and check isEmpty(). It returns true for a new list.
4. Append the elements Apple, Banana, Cherry and Banana using add(E e) and display the list.
5. Insert "Mango" at index 1 using add(int index, E element) and display the list.
6. Read the element at index 2 using get(int index).
7. Replace the element at index 0 with "Apricot" using set(int index, E element). It returns the old element.
8. Search the list:
 - contains() checks whether an element is present.
 - indexOf() gives the index of the first occurrence of "Banana".
 - lastIndexOf() gives the index of its last occurrence.
9. Display the number of elements using size() and check isEmpty() again.
10. Sort the list in ascending order using sort(Comparator.naturalOrder()), then in descending order using sort(Comparator.reverseOrder()), displaying the list after each sort.
11. Remove the element at index 0 using remove(int index). It returns the removed element.
12. Remove the first occurrence of "Banana" using remove(Object o). It returns true if the element was found.
13. Remove all elements using clear() and display size() and isEmpty().
14. Stop.

UML Diagram

<<interface>>

Collection<E>



```
<<interface>>
```

List<E>



```

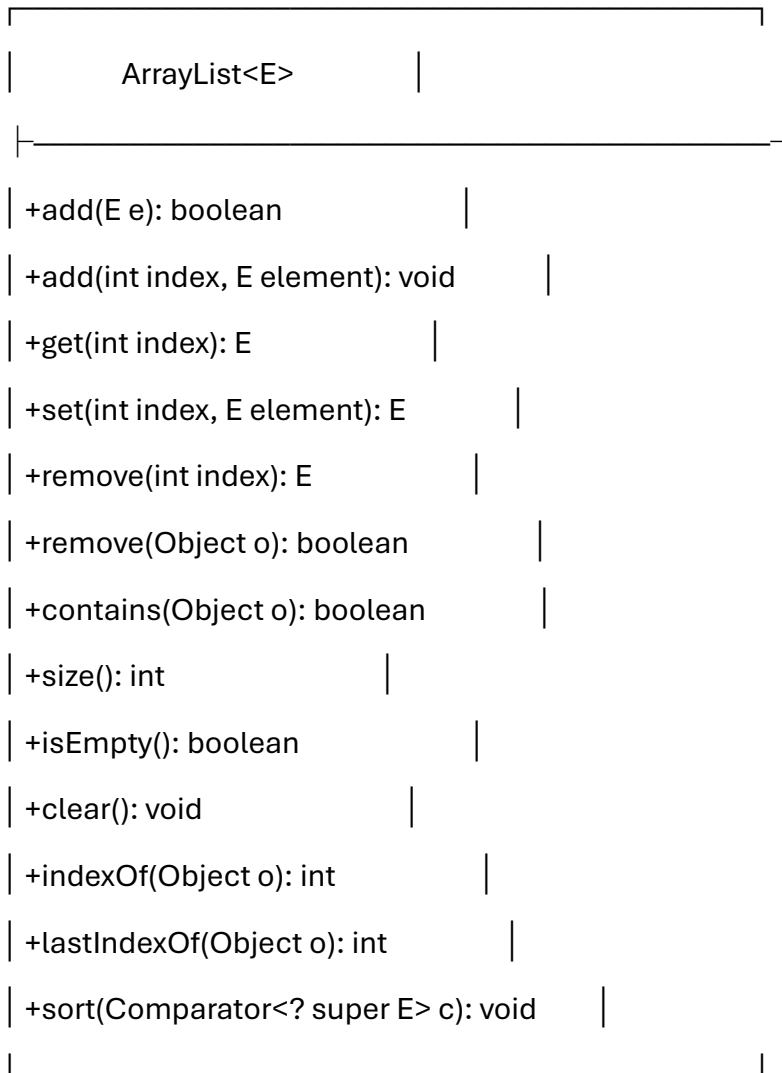
| (implements)

```

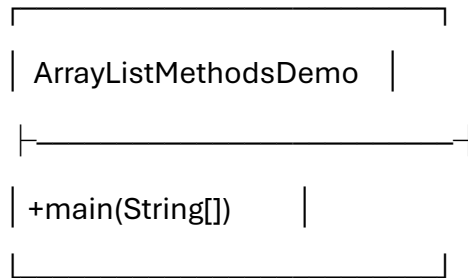
AbstractList<E>



| (extends)



:(uses)



`ArrayList` also implements `RandomAccess`, `Cloneable`, `Serializable`

Program:

```
package Observations;
```

```
import java.util.*;
```

```
public class ArrayListInterface {
```

```
    public static void main(String[] args) {
```

```
        ArrayList<String> list = new ArrayList<>();
```

```
        System.out.println("isEmpty() (new list) : " + list.isEmpty());
```

```
        // add(E e)
```

```
        list.add("Apple");
```

```
        list.add("Banana");
```

```
        list.add("Cherry");
```

```
        list.add("Banana");
```

```
        System.out.println("After add(E e) : " + list);
```

```
        // add(int index, E element)
```

```
        list.add(1, "Mango");
```

```
        System.out.println("After add(1, \"Mango\") : " + list);
```

```
// get()

System.out.println("get(2)          : " + list.get(2));


// set()

System.out.println("set(0, \"Apricot\")   : " + list.set(0, "Apricot"));
System.out.println("After set()        : " + list);


// contains(), indexOf(), lastIndexOf()

System.out.println("contains(\"Cherry\")   : " + list.contains("Cherry"));
System.out.println("contains(\"Grapes\")   : " + list.contains("Grapes"));
System.out.println("indexOf(\"Banana\")    : " + list.indexOf("Banana"));
System.out.println("lastIndexOf(\"Banana\") : " + list.lastIndexOf("Banana"));


// size() and isEmpty()

System.out.println("size()              : " + list.size());
System.out.println("isEmpty()           : " + list.isEmpty());


// sort()

list.sort(Comparator.naturalOrder());

System.out.println("Sorted (ascending)   : " + list);

list.sort(Comparator.reverseOrder());

System.out.println("Sorted (descending) : " + list);


// remove(int index) and remove(Object o)

System.out.println("remove(0)           : " + list.remove(0));

System.out.println("remove(\"Banana\")   : " + list.remove("Banana"));

System.out.println("After remove()      : " + list);

System.out.println("size()              : " + list.size());
```

```

// clear()

list.clear();

System.out.println("After clear()      : " + list);

System.out.println("size()            : " + list.size());

System.out.println("isEmpty()         : " + list.isEmpty());

}

}

```

Output:

```

PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java> & "C:\Program Files\Java\jdk-11.0.2\bin\java.exe" "-cp" "C:\Users\Sahithi_pyla\AppData\Roaming\Code\User\workspaceStorage\1\workspace\ArrayListInterface\ArrayListInterface.jar" ArrayListInterface
isEmpty() (new list)      : true
After add(E e)            : [Apple, Banana, Cherry, Banana]
After add(1, "Mango")     : [Apple, Mango, Banana, Cherry, Banana]
get(2)                   : Banana
set(0, "Apricot")         : Apple
After set()               : [Apricot, Mango, Banana, Cherry, Banana]
contains("Cherry")        : true
contains("Grapes")        : false
indexOf("Banana")         : 2
lastIndexOf("Banana")     : 4
size()                   : 5
isEmpty()                 : false
Sorted (ascending)       : [Apricot, Banana, Banana, Cherry, Mango]
Sorted (descending)      : [Mango, Cherry, Banana, Banana, Apricot]
remove(0)                 : Mango
remove("Banana")         : true
After remove()            : [Cherry, Banana, Apricot]
size()                   : 3
After clear()             : []
size()                   : 0
isEmpty()                 : true
PS C:\Users\Sahithi_pyla\Desktop\Sahithi_java>

```

Result:

Thus, the Java program demonstrating the ArrayList class methods was executed successfully and the output was verified.

- `add(E e)` appends to the end, while `add(int, E)` inserts at a position and shifts the later elements to the right.
- `set()` replaces an element and returns the old one, and `get()` reads an element without changing the list.
- `remove(int)` returns the removed element, while `remove(Object)` removes only the first match and returns true or false.
- `indexOf()` and `lastIndexOf()` return -1 if the element is not found.
- `sort()` sorts the list in place using the given Comparator.

In a list of Integer values, `list.remove(1)` removes the element at index 1, not the value 1. To remove the value, use `list.remove(Integer.valueOf(1))`. Also, `get()`, `set()` and `add(int, E)` throw `IndexOutOfBoundsException` if the index is out of range.